

MATH-150

The majority of the credit you receive will be based on the clarity of your ability to express your written answers and your work shown.

Turn your written portion in on a *separate* piece of paper.

Include a separate paper with your work. Show *all* your work in order to receive credit.

Save Yourself From Zombies!

There has been an outbreak of zombies at the University of San Diego. You, fortunately, have used your calculus skills to survive the outbreak. But you realize not everyone you know and love has the necessary calculus skills to survive. You team up with 5 friends and together try to survive the apocalypse. This is what we know about zombies ... they have the following weaknesses [1]:

- They wander around in an awkward pattern that can be described by an equation.
- They are not very clever. Thus, if you get into a fight with a zombie, even though they are strong, they will not be clever enough to pick up a weapon or guard themselves.
- Zombies cannot swim.
- Zombies are addicted to Snickers. This isn't a well known fact, but who isn't addicted to Snickers?

Between now and Thanksgiving, you will encounter zombies 4 times. As a Math 150 student, it is your duty to use your calculus skills to evade the zombies and help your friends survive. You must write up a plan for evading the zombies (you will be provided with a rubric for detailed expectations) and turn it into the professor by the due date listed. If you successfully evaded the zombies, your professor will give you the details of the next encounter. If you do not successfully evade the zombies, then one of your friends has been caught. Your professor will return your plan to you so you can try again. You must complete it correctly and turn it in with enough time to receive the next zombie challenge, and complete it by it's due date. After all four encounters, you will write a memoir of your experience evading zombies. Your memoir will include the details about the mathematics you used to evade zombies. It will be useful to future USD students when the next Zombie Apocalypse comes!

Future Zombie Encounters

1. Walking a tangential path. Due November 15.
3. Trapping the zombies in water. Due November 21.
2. Using a weapon against the zombies. Due November 30.
4. Finding safety. Due November 30.

The final memoir is due Tuesday, December 5.

References

- [1] Offutt, J, "How to Kill Monsters Using Common Household Items" 2015, Permuted Press